

DPT2

Physics: Electricity and Magnetism

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

Method © 2026 Erik Daniel Lindquist — Course Mastery Coach, armored drill. Items drafted with AI assistance under his direction.

How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word _____	The one fact that decides _____	Answer _____	[] sure [] unsure [] guess
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Items

1.

Two small copper spheres polished to a mirror finish carry fixed charges and repel with a force of 12 N. The lab is kept at 22 °C. If the distance between them is tripled, what is the new force?

- A. 108 N
- B. 4 N
- C. 12 N
- D. About 1.33 N

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2. [TRANSFER — reversed stem / different variable]

The force between two point charges is 8 N. One charge is doubled and the distance is halved. What is the new force?

- A. 16 N
- B. 64 N
- C. 32 N
- D. 8 N

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

A +2 μC test charge with a mass of 0.5 g feels a force of 0.06 N toward the east. The measurement was repeated five times. What is the electric field at that point?

- A. 30,000 N/C west
- B. 30,000 N/C east
- C. 0.03 N/C east
- D. 1.2×10^{-7} N/C east

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

A 12 V battery taken from a golf cart is connected to a 4 Ω resistor and a 2 Ω resistor in series. The wires are 30 cm long. What current flows?

- A. 72 A
- B. 2 A
- C. 9 A
- D. 6 A

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5. [TRANSFER — reversed stem / different variable]

Two identical resistors connected in parallel have an equivalent resistance of 5 Ω . What is the resistance of each?

- A. 10 Ω
- B. 5 Ω
- C. 20 Ω
- D. 2.5 Ω

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

A 60 W bulb installed in a hallway in 1995 runs on 120 V and stays on for 8 hours a day. What current does it draw?

- A. 480 Wh
- B. 2 A
- C. 0.5 A
- D. 240 Ω

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

A proton from a hospital cyclotron built in 2008 moves at 3×10^6 m/s parallel to a uniform 0.5 T magnetic field. What magnetic force acts on it?

- A. A force pointing along the field lines.
- B. 2.4×10^{-13} N
- C. Zero.
- D. A force opposite its motion that slows it down.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

A 200-turn coil of copper wire wrapped on a cardboard tube sees its magnetic flux drop from 0.05 Wb to 0.01 Wb in 0.2 s. The coil has a resistance of $8\ \Omega$. What EMF is induced?

- A. 1.6 V
- B. 60 V
- C. 0.2 V
- D. 40 V

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9. [TRANSFER — reversed stem / different variable]

A strong bar magnet is held perfectly still inside a coil. What EMF is induced in the coil?

- A. Zero.
- B. The maximum possible, since the flux is largest.
- C. It depends only on the number of turns.
- D. A steady nonzero EMF.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

An FM radio station that broadcasts with 50,000 watts transmits at 100 MHz. The tower is 150 m tall. What is the wavelength of its signal?

- A. It depends on the station's power.
- B. 300 m
- C. 3 m
- D. 3×10^{16} m

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

A transformer inside a phone charger with a USB-C port has 400 turns on the primary and 40 on the secondary. The primary is connected to 120 V. The charger case is white plastic. What is the secondary voltage?

- A. 12 V
- B. 1,200 V
- C. 120 V
- D. 3 V

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12. [TRANSFER — reversed stem / different variable]

An ideal step-down transformer draws 0.5 A at 120 V on the primary side and supplies 12 V on the secondary. What is the secondary current?

- A.** 60 A
- B.** 0.5 A
- C.** 0.05 A
- D.** 5 A

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer D Concept 1

Two small copper spheres ~~polished to a mirror finish~~ carry fixed charges and repel with a force of 12 N. ~~The lab is kept at 22 °C.~~ If the distance between them is tripled, what is the new force?

Target word: force after changing distance **Decides:** Force scales by $1/3^2 = 1/9$.

A	Inverted operation	Multiplies by 9; greater distance weakens the force.
B	Dropped element	Divides by 3, forgetting to square the distance factor.
C	Confusion set	Assumes force depends only on the charges.
D	Correct	$12 \div 9 \approx 1.33$ N.

2. Answer B Concept 1 · transfer

The force between two point charges is 8 N. One charge is doubled and the distance is halved. What is the new force?

Target word: combine two changes **Decides:** $\times 2$ for the charge, $\times 4$ for the halved distance.

A	Dropped element	Accounts for the charge but not the distance.
B	Correct	$8 \times 2 \times 4 = 64$ N.
C	Confusion set	Treats distance as $1/r$, giving $\times 2$ instead of $\times 4$.
D	Inverted operation	Assumes the two changes cancel, but halving r increases force.

3. Answer B Concept 2

A $+2 \mu\text{C}$ test charge ~~with a mass of 0.5 g~~ feels a force of 0.06 N toward the east. ~~The measurement was repeated five times.~~ What is the electric field at that point?

Target word: field magnitude and direction **Decides:** $E = 0.06 / (2 \times 10^{-6})$.

A	Confusion set	Reverses direction; the field points along the force on a positive charge.
B	Correct	$0.06 \div 0.000002 = 3.0 \times 10^4$ N/C, in the force direction for a positive charge.
C	Dropped element	Drops the micro prefix and divides by 2.
D	Inverted operation	Multiplies F by q instead of dividing.

4. Answer B Concept 3

A 12 V battery ~~taken from a golf cart~~ is connected to a 4Ω resistor and a 2Ω resistor in series. ~~The wires are 30 cm long.~~ What current flows?

Target word: series current **Decides:** $R = 4 + 2 = 6 \Omega$.

A	Inverted operation	Multiplies voltage by resistance.
B	Correct	$12 \text{ V} \div 6 \Omega = 2 \text{ A}$.
C	Confusion set	Uses the parallel rule ($4/3 \Omega$).
D	Dropped element	Uses only the 2Ω resistor.

5. Answer A Concept 3 · transfer

Two identical resistors connected in parallel have an equivalent resistance of $5\ \Omega$. What is the resistance of each?

Target word: solve for one resistor **Decides:** Two equal resistors in parallel give $R/2$.

A	Correct	$R/2 = 5$, so $R = 10\ \Omega$.
B	Dropped element	Treats the equivalent as each resistor's value.
C	Confusion set	Assumes parallel divides by 4.
D	Inverted operation	Halves 5 instead of doubling it.

6. Answer C Concept 4

A 60 W bulb ~~installed in a hallway in 1995~~ runs on 120 V ~~and stays on for 8 hours a day~~. What current does it draw?

Target word: current from power **Decides:** $I = P / V$.

A	Camouflage	Uses the hours to find daily energy, which was not asked.
B	Inverted operation	Divides V by P.
C	Correct	$60\ \text{W} \div 120\ \text{V} = 0.5\ \text{A}$.
D	Adjacent category	V^2/P is the bulb's resistance, not its current.

7. Answer C Concept 5

A proton ~~from a hospital cyclotron built in 2008~~ moves at $3 \times 10^6\ \text{m/s}$ parallel to a uniform 0.5 T magnetic field. What magnetic force acts on it?

Target word: magnetic force at $\theta = 0^\circ$ **Decides:** $\sin 0^\circ = 0$.

A	Adjacent category	That describes electric fields; magnetic force is perpendicular to B.
B	Dropped element	qvB with the angle factor left out.
C	Correct	$F = qvB \sin 0^\circ = 0$ for motion along the field.
D	Confusion set	Magnetic force is perpendicular to velocity and never acts like friction.

8. Answer D Concept 6

A 200-turn coil of copper wire ~~wrapped on a cardboard tube~~ sees its magnetic flux drop from 0.05 Wb to 0.01 Wb in 0.2 s. ~~The coil has a resistance of 8 Ω~~ . What EMF is induced?

Target word: induced EMF **Decides:** $\Delta\Phi = 0.04\ \text{Wb}$ over 0.2 s, times 200 turns.

A	Inverted operation	Multiplies by the time instead of dividing.
B	Wrong stem number	Adds the fluxes (0.06) instead of taking the change.
C	Dropped element	Leaves out the 200 turns.
D	Correct	$200 \times 0.04 \div 0.2 = 40\ \text{V}$.

9. Answer A Concept 6 · transfer

A strong bar magnet is held perfectly still inside a coil. What EMF is induced in the coil?

Target word: EMF with no change in flux **Decides:** Flux is not changing.

A	Correct	EMF requires changing flux; a still magnet gives constant flux.
B	Confusion set	EMF depends on the rate of change, not the size of the flux.
C	Dropped element	Turns multiply $\Delta\Phi/\Delta t$, which is zero here.
D	Adjacent category	Describes a battery, not induction.

10. Answer C Concept 7

An FM radio station ~~that broadcasts with 50,000 watts~~ transmits at 100 MHz. ~~The tower is 150 m tall.~~ What is the wavelength of its signal?

Target word: wavelength **Decides:** $\lambda = (3 \times 10^8) \div (1 \times 10^8)$.

A	Camouflage	Power affects strength, not wavelength.
B	Dropped element	Uses 1 MHz, dropping the factor of 100.
C	Correct	$3 \times 10^8 \div 10^8 = 3$ m.
D	Inverted operation	Multiplies c by f.

11. Answer A Concept 8

A transformer inside a phone charger ~~with a USB-C port~~ has 400 turns on the primary and 40 on the secondary. The primary is connected to 120 V. ~~The charger case is white plastic.~~ What is the secondary voltage?

Target word: secondary voltage **Decides:** $V_s = 120 \times 40 / 400$.

A	Correct	A 10:1 step-down gives 12 V.
B	Inverted operation	Inverts the turns ratio.
C	Dropped element	Ignores the turns ratio.
D	Wrong stem number	$120 \div 40$ uses the secondary turns alone.

12. Answer D Concept 8 · transfer

An ideal step-down transformer draws 0.5 A at 120 V on the primary side and supplies 12 V on the secondary. What is the secondary current?

Target word: solve for secondary current **Decides:** Power in = power out: 60 W.

A	Confusion set	60 is the power in watts, not a current.
B	Dropped element	Assumes current is unchanged.
C	Inverted operation	Current scales opposite to voltage, not with it.
D	Correct	$60 \text{ W} \div 12 \text{ V} = 5 \text{ A}$.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	11	
Wrong stem number	A figure copied from the prompt that is not the answer.	2	
Adjacent category	The right idea in the wrong bucket.	3	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	10	
Absolute language	Always, never, guarantee — where judgment applies.	0	
Confusion set	Every option is real; the test is exact role-matching.	8	
Camouflage	Answers a stem fact that was never the question.	2	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Coulomb's law is inverse-square	$F = kq_1q_2 / r^2$; force is proportional to each charge and to $1/r^2$.	Scaling force by $1/r$ instead of $1/r^2$.	"The distance is doubled / tripled / halved."
2	Electric field is force per charge	$E = F / q$, pointing in the direction of the force on a positive test charge; units N/C.	Multiplying F by q , dropping the micro prefix, or reversing direction.	A test charge, a measured force, and "field strength."
3	Ohm's law with series and parallel resistors	Series: $R = R_1 + R_2$; parallel: $1/R = 1/R_1 + 1/R_2$; then $I = V / R$.	Using the parallel rule for series resistors, or using one resistor alone.	A battery and two or more resistors.
4	Electric power	$P = IV = I^2R = V^2/R$, so $I = P / V$.	Inverting P / V , or reporting resistance or energy when current is asked.	A device's wattage and voltage.
5	Magnetic force on a moving charge	$F = qvB \sin \theta$, perpendicular to both v and B ; a charge moving parallel to B feels no force.	Ignoring the angle, or thinking the force points along the field.	A charge moving in a magnetic field at a stated angle.
6	Faraday's law of induction	$EMF = N \cdot \Delta\Phi / \Delta t$; an EMF is induced only while magnetic flux is changing.	Forgetting the number of turns, or expecting EMF from a steady strong field.	A coil, a flux change, and a time interval.
7	Electromagnetic waves: $c = f\lambda$	All EM waves travel at $c = 3.0 \times 10^8$ m/s in a vacuum, so $\lambda = c / f$.	Multiplying c by f , or thinking power or brightness changes wavelength.	A frequency in Hz, kHz, MHz, or GHz and "wavelength."
8	Ideal transformers	$V_s / V_p = N_s / N_p$, and power is conserved, so current changes inversely with voltage.	Inverting the turns ratio, or assuming current changes in the same direction as voltage.	Primary and secondary turns or voltages.

Course scope (public description): Physics: Electricity and Magnetism addresses principles related to the physics of electricity and magnetism. Students will study electric and magnetic forces and then apply that knowledge to the study of circuits with resistors and electromagnetic induction and waves. This course will focus on such topics as electric charge and electric field, electric currents and resistance, magnetism, electromagnetic induction and Faraday's law, and Maxwell's equation and electromagnetic waves.